



Qualification Pack

Milk Testing Facility Supervisor

QP Code: CUTM/FIC/Q0101

Version: 1.0

NSQF Level: 5

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CUTM/FIC/Q0101: Milk Testing Facility Supervisor

Brief Job Description

The Milk Testing Facility Supervisor should be able to collect market information for business planning and decision making, set-up paper recycling plant, operate the plant, manage the team, finance and administrative activities along with marketing, making quality products ensuring organizational health, safety and security.

Personal Attributes

A Milk Testing Facility Supervisor should be detail-oriented, disciplined, and knowledgeable in dairy quality standards. They must possess strong analytical skills, integrity, and technical proficiency in using milk testing equipment. Effective communication, documentation ability, and a commitment to hygiene, accuracy, and food safety are essential in this supervisory role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/FIC/N0101: Incoming milk quality check.](#)
2. [CUTM/FIC/N0102: In-process and outgoing milk quality check.](#)
3. [CUTM/FIC/N0103: Maintenance of milk testing equipment \(milk analyzer\)](#)
4. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Food Processing
Sub-Sector	Milk Quality checking
Occupation	Food Product Testing
Country	India
NSQF Level	5
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (Agriculture/Dairy technology/Dairy Science/Dairy Product Management and other relevant fields.) with NA of experience OR Completed 3-year diploma (after 10th) (Dairy technology/Dairy Science/Dairy Product Management and other relevant fields) with 1 Year of experience relevant experience OR 12th grade Pass with 2 Years of experience relevant fields experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant field experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant field experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	29/11/2026
NSQC Approval Date	30/11/2023
Version	1.0
Reference code on NQR	QG-05-FI-01396-2023-V1-CUTM
NQR Version	1.0

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CUTM/FIC/N0101: Incoming milk quality check.

Description

The NOS "Incoming Milk Quality Check" covers the procedures for assessing raw milk quality upon arrival at the facility. It includes sampling, testing for adulterants, analyzing key parameters like fat, SNF, and microbial load, ensuring compliance with regulatory standards, documenting results, and making decisions on acceptance or rejection based on quality standards.

Scope

The scope covers the following :

- The NOS "Incoming Milk Quality Check" includes verifying milk quality through sampling, physical and chemical analysis, and microbiological testing. It covers detecting adulterants, ensuring compliance with industry standards, maintaining hygiene during handling, documenting test results, and coordinating with procurement and quality teams for acceptance or rejection of milk consignments.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** analyzing quality of milk organoleptic ally by its flavor, odour, and appearance.
- PC2.** discard milk sample in case it is giving any foul, abnormal smell or have abnormal color.
- PC3.** blend 5 ml milk sample in test tube and add 5 ml ethanol and 4 drops of rosolic acid solution. Analyze color, rose red color indicates presence of neutralizer in milk, whereas brownish color indicates pure milk.
- PC4.** add few drops of 1% iodine solution in 10 ml of milk, to analyze the presence of starch. Development of blue color indicates presence of starch, whereas pure milk will not change the color
- PC5.** add 0.1 g resorcinol powder along with 10 ml HCl in 10 ml of milk, for added sugar detection. Heat milk in water bath for 5 min, color changes to red color indicates presence of sugar in milk.
- PC6.** verify fat, SNF and added water by using milk analyzer instrument.
- PC7.** ensure to heat the milk to 40oC before testing and then cool to 20oC. After thorough mixing take 20 ml of sample in the sampler.
- PC8.** validate fat content by Gerber method or using a milk analyzer.
- PC9.** verify SNF (Solid not fat) content by using lactometer or milk analyzer.
- PC10.** check acidity of milk by titrating 10 ml of milk with 0.1 N NaOH using 3 drops of phenolphthalein as indicator.
- PC11.** check COB (clot on boiling) test for heat stability of milk.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Composition and properties of milk, including fat content, protein levels, SNF (Solids-Not-Fat), and moisture.
- KU2.** Standard milk quality parameters such as acidity, density, temperature, and microbial load.
- KU3.** Various contaminants and adulterants in milk, including chemical, bacterial, and physical impurities.
- KU4.** Methods and equipment used for milk quality testing, such as lactometers, milk analyzers, and pH meters.
- KU5.** National and international quality standards for raw milk, such as FSSAI, BIS, and Codex guidelines.
- KU6.** Proper sampling techniques to ensure representative and contamination-free milk testing.
- KU7.** Organizational protocols for handling and storing incoming milk to maintain freshness and hygiene.
- KU8.** Importance of traceability and record-keeping for milk batches, supplier details, and test results.
- KU9.** Hygiene and safety procedures to prevent contamination during milk reception and testing.
- KU10.** Communication protocols for reporting deviations in milk quality to supervisors and suppliers.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret standard milk quality parameters, test reports, and regulatory guidelines.
- GS2.** Communicate test results clearly to suppliers, procurement teams, and supervisors.
- GS3.** Identify quality deviations and make decisions regarding acceptance or rejection of incoming milk.
- GS4.** Plan and organize sampling, testing, and documentation efficiently to ensure smooth workflow.
- GS5.** Apply problem-solving skills to address issues related to contamination, adulteration, or equipment malfunctions.
- GS6.** Use analytical thinking to interpret test results and determine necessary corrective actions.
- GS7.** Maintain accurate records of test results, supplier details, and milk batch traceability for compliance.
- GS8.** Follow organizational protocols and industry best practices for hygiene and safety during testing.
- GS9.** Work collaboratively with procurement, quality assurance, and production teams to ensure consistent milk quality.
- GS10.** Adapt to new testing technologies, software, and regulatory updates to improve efficiency and compliance.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. analyzing quality of milk organoleptic ally by its flavor, odour, and appearance.	3	6	-	2
PC2. discard milk sample in case it is giving any foul, abnormal smell or have abnormal color.	3	5	-	2
PC3. blend 5 ml milk sample in test tube and add 5 ml ethanol and 4 drops of rosolic acid solution. Analyze color, rose red color indicates presence of neutralizer in milk, whereas brownish color indicates pure milk.	3	5	-	2
PC4. add few drops of 1% iodine solution in 10 ml of milk, to analyze the presence of starch. Development of blue color indicates presence of starch, whereas pure milk will not change the color	3	3	-	2
PC5. add 0.1 g resorcinol powder along with 10 ml HCl in 10 ml of milk, for added sugar detection. Heat milk in water bath for 5 min, color changes to red color indicates presence of sugar in milk.	3	3	-	4
PC6. verify fat, SNF and added water by using milk analyzer instrument.	3	3	-	3
PC7. ensure to heat the milk to 40oC before testing and then cool to 20oC. After thorough mixing take 20 ml of sample in the sampler.	3	3	-	3
PC8. validate fat content by Gerber method or using a milk analyzer.	3	3	-	3
PC9. verify SNF (Solid not fat) content by using lactometer or milk analyzer.	2	3	-	3
PC10. check acidity of milk by titrating 10 ml of milk with 0.1 N NaOH using 3 drops of phenolphthalein as indicator.	2	3	-	3
PC11. check COB (clot on boiling) test for heat stability of milk.	2	3	-	3



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/FIC/N0101
NOS Name	Incoming milk quality check.
Sector	Food Processing
Sub-Sector	
Occupation	Food Product Testing
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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CUTM/FIC/N0102: In-process and outgoing milk quality check.

Description

The NOS "In-Process and Outgoing Milk Quality Check" covers monitoring milk quality during processing and final product inspection. It includes testing key parameters, detecting deviations, ensuring compliance with regulatory standards, maintaining hygiene, documenting results, and coordinating with production and quality teams to ensure consistent, safe, and high-quality milk products.

Scope

The scope covers the following :

- The NOS "In-Process and Outgoing Milk Quality Check" includes monitoring milk quality at different processing stages, conducting chemical, physical, and microbiological tests, detecting deviations, ensuring compliance with industry standards, maintaining hygiene protocols, documenting test results, and collaborating with production and quality assurance teams to ensure safe and high-quality milk products.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** check presence or absence of alkaline phosphatase enzyme in milk by phosphatase test.
- PC2.** transfer 5 ml of buffer substrate solution (mixture of disodium p- nitrophenyl phosphate and carbonate- bicarbonate buffer) in test tube and place in water bath maintained at 37 degrees C.
- PC3.** ensure to mix content thoroughly after 5 min by adding 1 ml of milk.
- PC4.** prepare blank from boiled milk and incubate all tubes in water bath at 37 degrees C for 2 hr. After 2hr remove tubes and compare colour using comparator and special disk.
- PC5.** verify tube containing raw or unpasteurized milk to show yellow colour whereas the boiled milk to remain white in colour
- PC6.** verify fat, SNF and added water by using milk analyzer instrument after pasteurization or by manually using Gerber method and using lactometer.
- PC7.** ensure to pump in the milk sample in sensing system to measure milk temperature. Milk type is selected by following the required steps. Required steps: Menu > Setting > Select milk type then arrow mark to select it. Press (*) key.
- PC8.** ensure to pump out milk sample, once count down timer stops and values of fat, SNF and added water are shown.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Key quality parameters of in-process and outgoing milk, including fat content, SNF, microbial load, acidity, and temperature.

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- KU2.** Various testing methods such as organoleptic, chemical, microbiological, and instrumental analysis for quality checks.
- KU3.** Role of pasteurization, homogenization, and other processing techniques in maintaining milk quality.
- KU4.** Standard operating procedures (SOPs) for in-process quality monitoring to ensure consistency.
- KU5.** National and international milk quality standards, such as FSSAI, BIS regulations.
- KU6.** Common defects and spoilage indicators in processed milk, including off-flavors, curdling, and contamination.
- KU7.** Proper documentation and traceability of quality control data for regulatory compliance and audits.
- KU8.** Organizational protocols for handling, packaging, and storage of processed milk to prevent deterioration.
- KU9.** Corrective actions and decision-making in case of non-conforming milk batches.
- KU10.** Communication and reporting procedures for coordinating with production, logistics, and regulatory teams.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret in-process and final milk quality parameters, test reports, and regulatory standards.
- GS2.** Communicate quality test results effectively to production, quality assurance, and logistics teams.
- GS3.** Identify deviations in milk quality and make informed decisions regarding corrective actions.
- GS4.** Plan and organize regular quality checks to ensure compliance with industry and organizational standards.
- GS5.** Use problem-solving skills to address issues such as microbial contamination, spoilage, or process inefficiencies.
- GS6.** Apply analytical thinking to assess test results and determine process adjustments for quality improvement.
- GS7.** Maintain accurate records of in-process and outgoing milk quality checks for traceability and audits.
- GS8.** Follow hygiene, safety, and regulatory protocols to prevent contamination and maintain product integrity.
- GS9.** Collaborate with production and packaging teams to ensure consistent quality before dispatch.
- GS10.** Adapt to new testing methods, equipment, and digital reporting systems to enhance efficiency and accuracy.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. check presence or absence of alkaline phosphatase enzyme in milk by phosphatase test.	5	8	-	-
PC2. transfer 5 ml of buffer substrate solution (mixture of disodium p- nitrophenyl phosphate and carbonate- bicarbonate buffer) in test tube and place in water bath maintained at 37 degrees C.	5	8	-	-
PC3. ensure to mix content thoroughly after 5 min by adding 1 ml of milk.	5	3	-	-
PC4. prepare blank from boiled milk and incubate all tubes in water bath at 37 degrees C for 2 hr. After 2hr remove tubes and compare colour using comparator and special disk.	5	8	-	-
PC5. verify tube containing raw or unpasteurized milk to show yellow colour whereas the boiled milk to remain white in colour	5	9	-	-
PC6. verify fat, SNF and added water by using milk analyzer instrument after pasteurization or by manually using Gerber method and using lactometer.	5	8	-	-
PC7. ensure to pump in the milk sample in sensing system to measure milk temperature. Milk type is selected by following the required steps. Required steps: Menu > Setting > Select milk type then arrow mark to select it. Press (*) key.	5	8	-	-
PC8. ensure to pump out milk sample, once count down timer stops and values of fat, SNF and added water are shown.	5	8	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/FIC/N0102
NOS Name	In-process and outgoing milk quality check.
Sector	Food Processing
Sub-Sector	
Occupation	Food Product Testing
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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CUTM/FIC/N0103: Maintenance of milk testing equipment (milk analyzer)

Description

The NOS "Maintenance of Milk Testing Equipment (Milk Analyzer)" covers routine cleaning, calibration, troubleshooting, and preventive maintenance of milk analyzers. It includes ensuring accurate test results, following safety protocols, documenting maintenance activities, identifying malfunctions, and coordinating with technical teams or service providers for repairs to maintain equipment efficiency and compliance.

Scope

The scope covers the following :

- The NOS "Maintenance of Milk Testing Equipment (Milk Analyzer)" includes routine cleaning, calibration, troubleshooting, and preventive maintenance of milk analyzers. It covers identifying technical issues, ensuring accurate test results, following safety and hygiene protocols, maintaining records, and coordinating with service providers for repairs to ensure optimal equipment performance and regulatory compliance.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** ensure milk sample should be homogeneous, free from air bubbles and temperature of milk should be 10 oC to 40oC.
- PC2.** ensure milk is cooled and some of fats gets separated. Heat the milk to 40 – 42oC in water bath slowly and cool it down to 20-25oC for dissolving separated fat.
- PC3.** assure to stir milk while heating and cooling by ultrasonic stirrer to remove air bubbles.
- PC4.** manage rinsing three times by five cycles of water pumped in and out by selecting appropriate option. Contaminated water is poured out Option : Menu > Cleaning > Rinse
- PC5.** manage daily cleaning process by using 2% alkaline solution, prepared by mixing thoroughly 490 ml water in 10 ml daily clean concentrate Option : Menu > Cleaning > Daily Clean
- PC6.** clean in and outlet pipes with fresh water by using sample container and then by alkaline solution and rinse thrice.
- PC7.** ensure to clean the in and out pipes with fresh water after alkaline wash.
- PC8.** facilitate weekly cleaning by using 10% acid solution, prepared by mixing thoroughly 50 ml weekly clean concentrate and 450 ml water Option: Menu > Cleaning > Weekly Clean
- PC9.** cleanse in and outlet pipes with fresh water by using sample container and then by acid weekly solution and rinse for 5 times.
- PC10.** ensure to clean in and out pipes with fresh water after the acid wash Option: click the Menu key to come out of cleaning procedure
- PC11.** carry out manual cleaning by switching off machine and keeping fresh water in sample stand. Remove rubber plug and insert syringe plunger instead.
- PC12.** pull up and down plunger several times in order to get out milk residues out. Repeat process 3 - 4 times.
- PC13.** keep container on surface and using plunger check that no water is present inside sensor system.

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- PC14.** remove plunger, insert plug and check with water for proper reading. in case pipe system have got blocked due to big milk solids, never apply more pressure, instead slow up and down plunger so that water movement is possible.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Working principles and components of milk analyzers, including sensors, calibration units, and display systems.
- KU2.** Standard operating procedures (SOPs) for routine maintenance, cleaning, and calibration of milk testing equipment.
- KU3.** Common technical issues in milk analyzers, such as sensor drift, clogging, and inaccurate readings.
- KU4.** Proper handling and storage of reagents, cleaning solutions, and consumables used in milk testing equipment.
- KU5.** National and international standards for milk analyzer performance and accuracy, including FSSAI and BIS guidelines.
- KU6.** Troubleshooting techniques to identify and rectify errors in milk testing readings.
- KU7.** Preventive maintenance schedules and record-keeping for ensuring equipment efficiency and compliance.
- KU8.** Safety precautions and best practices for handling electronic and chemical components in milk analyzers.
- KU9.** Importance of software updates and firmware maintenance for digital milk analyzers.
- KU10.** Organizational protocols for coordinating with service providers and vendors for repairs and technical support.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical manuals, calibration guidelines, and equipment specifications for milk analyzers.
- GS2.** Communicate maintenance schedules, issues, and troubleshooting steps effectively to the technical team and supervisors.
- GS3.** Identify and diagnose common technical issues in milk analyzers, such as sensor errors, inaccurate readings, and clogging.
- GS4.** Plan and organize routine maintenance, calibration, and cleaning schedules to ensure accurate test results.
- GS5.** Apply problem-solving skills to resolve equipment malfunctions and minimize downtime.
- GS6.** Use analytical thinking to assess equipment performance trends and take preventive actions.
- GS7.** Maintain accurate records of maintenance activities, calibration data, and equipment performance logs.



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- GS8.** Follow safety and hygiene protocols while handling electronic components, reagents, and cleaning solutions.
- GS9.** Collaborate with equipment vendors and service providers for timely repairs and technical support.
- GS10.** Stay updated on advancements in milk testing technologies and implement best practices for equipment maintenance.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. ensure milk sample should be homogeneous, free from air bubbles and temperature of milk should be 10 oC to 40oC.	3	5	-	-
PC2. ensure milk is cooled and some of fats gets separated. Heat the milk to 40 – 42oC in water bath slowly and cool it down to 20-25oC for dissolving separated fat.	3	5	-	-
PC3. assure to stir milk while heating and cooling by ultrasonic stirrer to remove air bubbles.	3	5	-	-
PC4. manage rinsing three times by five cycles of water pumped in and out by selecting appropriate option. Contaminated water is poured out Option : Menu > Cleaning > Rinse	3	5	-	-
PC5. manage daily cleaning process by using 2% alkaline solution, prepared by mixing thoroughly 490 ml water in 10 ml daily clean concentrate Option : Menu > Cleaning > Daily Clean	3	5	-	-
PC6. clean in and outlet pipes with fresh water by using sample container and then by alkaline solution and rinse thrice.	3	5	-	-
PC7. ensure to clean the in and out pipes with fresh water after alkaline wash.	3	5	-	-
PC8. facilitate weekly cleaning by using 10% acid solution, prepared by mixing thoroughly 50 ml weekly clean concentrate and 450 ml water Option: Menu > Cleaning > Weekly Clean	3	5	-	-
PC9. cleanse in and outlet pipes with fresh water by using sample container and then by acid weekly solution and rinse for 5 times.	3	4	-	-
PC10. ensure to clean in and out pipes with fresh water after the acid wash Option: click the Menu key to come out of cleaning procedure	3	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. carry out manual cleaning by switching off machine and keeping fresh water in sample stand. Remove rubber plug and insert syringe plunger instead.	3	4	-	-
PC12. pull up and down plunger several times in order to get out milk residues out. Repeat process 3 - 4 times.	3	4	-	-
PC13. keep container on surface and using plunger check that no water is present inside sensor system.	2	2	-	-
PC14. remove plunger, insert plug and check with water for proper reading. in case pipe system have got blocked due to big milk solids, never apply more pressure, instead slow up and down plunger so that water movement is possible.	2	2	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/FIC/N0103
NOS Name	Maintenance of milk testing equipment (milk analyzer)
Sector	Food Processing
Sub-Sector	
Occupation	Food Product Testing
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by AB
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding
- Online proctoring to be done

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 70% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

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Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/FIC/N0101.Incoming milk quality check.	30	40	-	30	100	25
CUTM/FIC/N0102.In-process and outgoing milk quality check.	40	60	-	-	100	25
CUTM/FIC/N0103.Maintenance of milk testing equipment (milk analyzer)	40	60	-	-	100	20
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	0	0	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
Total	160	260	0	0	450	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.